# MAHER: Multi-Agent Hierarchical Expert Re-identification for Large-Scale Wildlife Recognition

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# Overview

**The problem:** identifying wild animals without physical tags is hard in the real world, because populations are dynamic — new individuals keep appearing that a model has never seen before.

**Our approach:** a two-stage hierarchical pipeline.

1. **Model A** (generalist) looks at a photo and predicts the *species*.
2. A **routing table** picks the matching **Model B** specialist for that species.
3. That specialist looks at the *same* photo and identifies the *individual* animal — or flags it as new.

**Scope:** 4 species — hyena, leopard, macaque, dog.

# Data & Key Decisions

- Source: the `wildlife-datasets` package — SpottedHyenaID109, LeopardID102, MacaqueFaces, DogFaceNet.
- **12,308** labelled images across the 4 species.
- Scoped down from WildlifeReID-10k's ∼33 species to 4, for an honours-project timeline and CPU-only compute.
- **Split strategy:** individual-level, not random — some individuals held out *entirely* (open-set / novel), the rest keep most images in train but hold out a few for test too (so closed-set accuracy is measurable as well).
- Caught and fixed a real bug: an early split version left *zero* test images belonging to a known individual, silently breaking closed-set evaluation.
- Reconciled two independently-built data pipelines (SQLite-based vs. a shared `metadata.csv`) that had diverged mid-project.

# Model A — Species Classifier

**Architecture**

- ResNet50, frozen ImageNet backbone
- Only the final linear head trains
- Cheap on CPU — fast to iterate

**Training**

- Cross-entropy loss, Adam optimiser
- 5 epochs

**Result**

**Test accuracy**

99.93%

4,069 test images — near-perfect, since the 4 species are visually very distinct.

Two independently-trained checkpoints exist (different split methodologies); the deployed version uses the identity-split checkpoint (100% validation accuracy).

# Model B — Individual Re-identification

**Architecture:** ViT-Tiny backbone + ArcFace loss, trained *separately per species* — one specialist per species, not one shared flat model. This species-conditioning is the core architectural contribution over a flat baseline.

**Inference:** cosine similarity against a gallery of training embeddings baked into each checkpoint; a similarity threshold decides known individual vs. *"new animal"* (open-set rejection).

| Species | AUC | TNR@95 | BAKS | Combined |
|---|---|---|---|---|
| macaque | 0.98 | 0.94 | 0.95 | **0.94** |
| dog | 0.92 | 0.51 | 0.92 | 0.68 |
| leopard | 0.74 | 0.15 | 0.65 | 0.30 |
| hyena | 0.72 | 0.16 | 0.51 | 0.30 |

Training data volume is the dominant factor, not species difficulty — macaque/dog had far more images than hyena/leopard.

# UI, Integration & Deployment

**Integration:** one photo in → Model A predicts species → routing table picks the matching Model B specialist → same photo → individual or *"New animal"*.

**Interface:** a public web app (Flask), built on the same inference code as the rest of the pipeline.

**Deployment:**

- Containerised with Docker.
- Served behind a reverse proxy with automatic HTTPS.
- Live, publicly accessible demo.

**Live demo:** `https://maher.lwazi.gay`